When Extrapolation Fails: Testing Conformal Prediction's Exchangeability Assumption Across Surrogate Architectures

Course: 23AID201

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Abstract

Conformal prediction's coverage guarantee depends on an exchangeability assumption between calibration and test data - it is not designed to hold under distribution shift. Gopakumar et al. (2026) name this as a second, untested limitation of conformal prediction (CP) for surrogate-model uncertainty quantification. We stress-test this assumption in a discrete-event ER simulation by pushing the test distribution's demand level up to 3x the training range, using two structurally different surrogate architectures - gradient boosting and a neural network (MLP). Coverage collapses in both cases, confirming the limitation, but the failure mechanism differs sharply: gradient boosting's predictions freeze at the training boundary, while the MLP keeps extrapolating a trend that overshoots the true value by 6-9x more error. The ability to extrapolate is not automatically an advantage - it depends on whether the true relationship continues the trend the model learned, or saturates, as it does here.

1. Introduction

Conformal prediction (CP) provides a distribution-free, finite-sample coverage guarantee, but that guarantee rests on an exchangeability assumption: calibration data and test data must be drawn from the same distribution. Gopakumar et al. (2026), who validate CP for surrogate-model uncertainty quantification in physics simulation domains, explicitly flag this assumption as untested under distribution shift. This report investigates what happens when it is violated in a discrete-event ER queueing simulation, and whether the failure mode depends on the surrogate model's architecture.

2. Method

Calibration was kept fixed exactly as established for in-distribution evaluation (arrival-rate multiplier in [0.8, 1.3] relative to the real calibrated rate). The test distribution's arrival-rate multiplier was then pushed progressively outward - 1.5x, 1.8x, 2.0x, 2.5x, 3.0x - an escalating demand surge well outside the training range, while staffing capacity remained drawn from its normal range, isolating the shift to demand alone. Coverage was measured at each severity level for both standard and Mondrian conformal prediction, using two surrogate architectures trained on identical data: a gradient-boosting regressor (GBR, tree-based) and a multilayer perceptron (MLP, a small neural network), which achieve near-identical in-distribution accuracy (R-squared within 0.01 of each other on every target), ensuring any difference observed under distribution shift reflects architecture, not overall model quality.

3. Results

Table 1 shows standard CP's coverage collapsing as severity increases past the training boundary (1.3x), for the gradient-boosting surrogate.

Arrival multiplier	n_patients	mean_wait	mean_total	p95_wait
1.3 (in range)	93.3%	83.0%	88.0%	82.0%
1.5	78.7%	68.7%	73.7%	69.7%
1.8	70.3%	42.7%	47.3%	43.0%
2.0	64.7%	34.3%	39.0%	33.0%
3.0	31.7%	12.3%	4.7%	78.3%*

Table 1. Gradient-boosting surrogate: coverage vs. arrival severity. *p95_wait's apparent recovery at 3.0x is not the interval working correctly - it is explained in Section 4.

Repeating the test with the MLP surrogate produces an even faster collapse for two of the four targets: n_patients and mean_total_minutes both reach exactly 0% coverage by 2.0x severity, compared to the gradient-boosting surrogate's more gradual decline to 31.7% and 4.7% respectively at 3.0x (Table 2).

Arrival multiplier	n_patients (MLP)	mean_total (MLP)	n_patients (GBR)	mean_total (GBR)
1.3 (in range)	93.0%	88.3%	93.3%	88.0%
1.8	6.7%	2.3%	70.3%	47.3%
2.0	0.0%	0.0%	64.7%	39.0%
3.0	0.0%	0.0%	31.7%	4.7%

Table 2. Standard CP coverage, MLP vs. gradient-boosting (GBR) surrogate, standard CP.

This is initially counter-intuitive: gradient-boosting (tree-based) models cannot extrapolate past the range of their training data - their predictions clip to whatever leaf the boundary training points fell into, effectively freezing. The MLP has no such limitation and its predictions keep changing as severity increases. One would expect the architecture capable of extrapolating to handle distribution shift better. It does not.

Comparing both models' predictions against the true simulated value at fixed capacity (30 servers) reveals why (Table 3). The true DES output saturates at extreme demand rather than growing proportionally: at extreme overload, more arriving patients simply queue up and do not complete service within the simulated day, so the count of completed visits levels off. The gradient-boosting model's frozen prediction happens to land close to this saturating value by coincidence - the true function actually is roughly flat in this regime, so "frozen" is qualitatively the right shape. The MLP instead extrapolates the upward trend it learned near the training boundary and continues it indefinitely, overshooting the true value by a large margin.

n_capacity = 30	Arrival 1.0x	Arrival 1.3x	Arrival 2.0x	Arrival 3.0x	n_patients, true
True value	235.2	244.8	258.6	280.9	(saturating)
GBR prediction	240.5	247.8	247.8 (frozen)	247.8 (frozen)	error ≈ 33
MLP prediction	234.8	245.8	336.5	521.2	error ≈ 240

Table 3. True vs. predicted n_patients at fixed capacity, both architectures. The MLP's error at 3.0x severity is roughly 7x larger than the gradient-boosting model's, despite - or rather, because of - its ability to extrapolate.

4. Discussion

The p95_wait_minutes target's apparent coverage "recovery" at extreme severity (Table 1, 3.0x) is a data-generating-process artifact, not a sign the interval is working. At extreme overload, fewer patients finish service within the simulated day at all, shrinking and changing the composition of the completed-visit pool the 95th percentile is computed over. The true value drifting back toward the frozen prediction is coincidental, driven by this composition shift, not by the surrogate or the interval becoming more reliable.

The central finding - that an architecture's ability to extrapolate is not automatically an advantage - depends entirely on whether the true relationship continues the trend a model learned near its training boundary. Here it does not: the relationship saturates due to a censoring mechanism in how the outcome is measured. A model that confidently extrapolates the wrong trend can be worse than one that cannot extrapolate at all. This is a meaningful caveat for any practitioner choosing a surrogate architecture specifically to "handle" out-of-distribution inputs: the choice should depend on the true relationship's expected shape beyond the training range, not on extrapolation capability alone.

One property is worth noting as a partial mitigation: the failure is detectable, not silent. Residuals grow and coverage measurably collapses rather than the surrogate confidently reporting a narrow, wrong interval with no signal that something is amiss. This does not restore the coverage guarantee, but it means the failure mode is one a monitoring system could in principle catch.

5. Conclusion

Conformal prediction's exchangeability assumption breaks down outside the training distribution in this domain, confirming Gopakumar et al.'s second stated limitation. Testing two structurally different surrogate architectures shows the breakdown is not an artifact of one architecture's specific limitation (tree-based extrapolation freezing) - it occurs regardless of architecture, though the failure mechanism and severity differ. Counter-intuitively, the architecture that cannot extrapolate at all degraded more gracefully than the one that could, because the true relationship in this domain saturates rather than growing without bound - a reminder that extrapolation capability is not inherently protective against distribution shift.

References

Gopakumar, et al. (2026). Conformal Prediction for surrogate-model uncertainty quantification in physics simulation domains (PDEs, MHD, weather, fusion). [Base paper]

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